



Exam Questions

How aircraft fly

Marcos Carrera Valle

Exam Questions

Who is credited with the first powered flight?

- a) The Wright Brothers
- b) Leonardo da Vinci
- c) Samuel Langley
- d) Otto Lilienthal

Which pioneer is known for studying glider flight and aerodynamics?

- a) Robert Goddard
- b) Otto Lilienthal
- c) Chuck Yeager
- d) John Glenn

Exam Questions

What was the key contribution of the Wright brothers to aviation?

- a) First supersonic flight
- b) First liquid-fueled rocket
- c) Controlled, sustained flight with three-axis control
- d) First manned spaceflight

What is the main effect of increasing the aspect ratio of a wing?

- a) Increases induced drag
- b) Increases wingtip vortices
- c) Reduces induced drag
- d) Increases maneuverability

Exam Questions

A high aspect ratio wing is best suited for:

- a) High-speed fighter jets
- b) Low-speed gliders
- c) Helicopters
- d) Supersonic aircraft

The chord line of an airfoil is:

- a) The distance from wingtip to wingtip
- b) The imaginary line connecting the leading and trailing edge
- c) The maximum camber of the airfoil
- d) The airfoil thickness

Exam Questions

Bernoulli's Principle states that an increase in velocity leads to:

- a) An increase in pressure
- b) A decrease in pressure
- c) No change in pressure
- d) A decrease in temperature

Why is Bernoulli's equation important in aerodynamics?

- a) It explains how airspeed changes around an airfoil
- b) It determines fuel efficiency
- c) It calculates stall speed
- d) It explains wingtip vortices

Exam Questions

What does the first digit in a NACA 4-digit airfoil code represent?

- a) Maximum thickness percentage
- b) Maximum camber percentage
- c) Position of maximum thickness
- d) Trailing edge shape

Which NACA airfoil has no camber?

- a) NACA 2412
- b) NACA 0012
- c) NACA 4415
- d) NACA 6321

Exam Questions

Increasing the camber of an airfoil generally:

- a) Increases maximum lift coefficient
- b) Reduces drag at high speeds
- c) Improves supersonic performance
- d) Reduces lift

What is the main purpose of flaps?

- a) Increase lift and drag at low speeds
- b) Reduce drag at high speeds
- c) Improve stability in supersonic flight
- d) Reduce wingtip vortices

Exam Questions

Leading-edge slats work by:

- a) Reducing the critical angle of attack
- b) Delaying flow separation
- c) Reducing lift coefficient
- d) Increasing wave drag

What happens when flaps are fully extended?

- a) Decrease lift and decrease drag
- b) Increase lift and increase drag
- c) Increase lift and decrease drag
- d) Decrease lift and increase drag

Exam Questions

Parasitic drag includes:

- a) Induced drag and vortex drag
- b) Skin friction and pressure drag
- c) Shock wave drag only
- d) Drag caused by wingtip vortices

Induced drag is highest when:

- a) The aircraft is flying at low speed and high angle of attack
- b) The aircraft is flying at high speed
- c) The aircraft is flying at sea level
- d) The aircraft has no lift

Exam Questions

What effect does streamlining an aircraft have?

- a) Reduces wave drag but increases skin friction
- b) Increases induced drag
- c) Reduces parasitic drag
- d) Increases lift

Jet engines are most efficient at:

- a) Low speeds and low altitudes
- b) High speeds and high altitudes
- c) High speeds and low altitudes
- d) Low speeds and high altitudes

Exam Questions

A propeller engine is typically used for:

- a) High-speed military jets
- b) Supersonic airliners
- c) Regional and general aviation aircraft
- d) Hypersonic travel

What type of thrust is produced by a turbojet engine?

- a) Piston thrust
- b) Pure jet thrust
- c) Propeller-driven thrust
- d) Chemical propulsion

Exam Questions

An aircraft is flying at 100 m/s in air with a density of 1.225 kg/m^3 . Its wing area is 20 m^2 , and its lift coefficient is 0.8. What is the lift force?

- a) 78,400 N
- b) 98,000 N
- c) 122,500 N
- d) 50,000 N

Exam Questions

If the lift coefficient of an aircraft is 1.2, and all other factors remain the same, how does the lift force change?

- a) Becomes zero
- b) Decreases
- c) Stays the same
- d) Increases

Exam Questions

An aircraft generates 150,000 N of lift while flying at 300 m/s. The air density is 0.3 kg/m^3 and its wing area is 30 m^2 . What is the lift coefficient?

- a) 0.93
- b) 1.44
- c) 1.26
- d) 0.37

Exam Questions

A fighter jet's wings have a total area of 40 m^2 , and its aspect ratio is 8. What is its wingspan?

- a) 10.6 m
- b) 15.3 m
- c) 17.9 m
- d) 25.0 m

Exam Questions

If the airspeed of an aircraft is 200 m/s, and it increases to 400 m/s, how does lift change?

- a) Increases 2×
- b) Increases 4×
- c) Decreases 2×
- d) Stays the same

Exam Questions

An aircraft with a drag coefficient of 0.02 has a lift-to-drag ratio of 10. What is its lift coefficient?

- a) 0.1
- b) 0.2
- c) 0.3
- d) 0.5

Exam Questions

If the air density at high altitude is 0.5 kg/m^3 , compared to 1.225 kg/m^3 at sea level, how does the lift change?

- a) Increases
- b) Decreases
- c) Stays the same
- d) Becomes zero

Exam Questions

If an aircraft's wing area is 60 m^2 , and it reduces to 30 m^2 , how does lift change?

- a) Reduces by half
- b) Doubles
- c) Stays the same
- d) Becomes zero

Exam Questions

A cargo plane's wingspan is 50 m, and its aspect ratio is 7. What is its wing area?

- a) 203.3 m²
- b) 306.4 m²
- c) 357.1 m²
- d) 408.8 m²

Exam Questions

If an aircraft increases its angle of attack too much, what happens?

- a) Lift increases indefinitely
- b) Drag disappears
- c) The aircraft stalls
- d) Thrust increases

Answers

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